Omsk 2400

Qu 2:

$2024s + 1265t = \gcd(2024, 1265)$

Find: $\gcd(2024, 1265)$

$2024 = 1265 \cdot 1 + 759$

$1265 = 759 \cdot 1 + 506$

$759 = 506 \cdot 1 + 253$

$506 = 253 \cdot 2 + 0$

$\gcd(2024, 1265) = 253$ by euclides alg...

Sub:

$253 = 759 - 1 \cdot 506$

$506 = 1265 - 1 \cdot 759$

$\Rightarrow 253 = 759 - 1(1265 - 1 \cdot 759) = 2 \cdot 759 - 1 \cdot 1265$

$759 = 2024 - 1 \cdot 1265$

$\Rightarrow 253 = 2(2024 - 1 \cdot 1265) \Rightarrow 253 =$

$\Rightarrow 2 \cdot 2024 - 3 \cdot 1265.$

Detta innebär att $s = 2$ och $t = -3$.

$2024(2) + 1265(-3) = 253$